

Web Programming

Chapter One:

The world Wide Web

Outline

Chapter 1: The world Wide Web (4hr)

- 1.Course overview
- 2.The Internet and World Wide Web
- 3.Web Hosting and Domain Name Registration

Basic Terminology in Web Technology

- A **network** is a **collection of devices** (such as computers, routers, switches, and servers) that are connected and can communicate with each other.
- Networks can be **wired or wireless**, and they use various protocols and technologies to facilitate the exchange of data and resources among the connected devices.
- The **Internet** is a **globally interconnected computer networks** that allows for the **exchange of information and communication** between devices.
- **On-line:-** “being on-line”; they are using the Internet.
- internet services
 - electronic mail
 - www
 - mobile apps
 - file sharing (FTP)
 - online telephony
 - Instant messaging (chat)
 - Remote login

World Wide Web (WWW)

The World Wide Web (WWW):- It is huge collection of webpages of information linked each other around the globe.

- It is a **global information system** that **allows users to access and share information, communicate, and perform various tasks** through interconnected web pages and websites.
- All public websites or web pages that people may access on their local computers and other devices through the internet are collectively known as the **World Wide Web** or W3.
- WWW is defined as the **collection of different websites around the world, containing different information shared via local servers**(or computers).
- The web is governed by the w3c (world wide web consortium) (www.w3.org)

World Wide Web

The **key components** of the World Wide Web include:

- **Web pages:** These are individual documents accessible through a web browser, containing text, images, videos, and other multimedia content.
- **Websites:** Collections of interconnected web pages, often focused on a particular topic, organization, or service.
- **URLs (Uniform Resource Locators):** The unique addresses used to access specific web pages or resources on the web.
- **HTTP (Hypertext Transfer Protocol):** The standard protocol used to transfer web pages and other resources over the internet.
- **Web browsers:** Software applications, such as Google Chrome, Mozilla Firefox, or Microsoft Edge, that allow users to access and navigate

Hypertext Transfer Protocol(HTTP)

- Is original web communication protocol which support connectionless communication between web server and its clients.
- This is the protocol being used to transfer hypertext documents that makes the World Wide Web possible.
- It defines how web browsers and web servers communicate with each other.
- HTTP uses a **request-response model**, where the client (web browser) sends a request to the server, and the server responds with the requested data.
- The HTTP is an application layer protocol that allows web based applications to communicate and exchange data.
- the HTTP is the **messenger of the web**.
- it is used to deliver contents, eg images, videos, audios, documents etc
- Facilitates the communication and defines the rule,syntax,semantic or formats of the communication.

Properties of Hypertext Transfer Protocol(HTTP)

1. HTTP is **connectionless** :-after making the request, the **client disconnect** from the server, then when the response is ready the **server re-establish the connection** again and deliver the response.
 2. the HTTP **can deliver any sort of data**, as long as the two computers are **able to read it**.
 3. The HTTP is a **Stateless**:- the client and the server know about each other **just during the current request**, if it closes, and the two computers want to connect again, they need to provide information to each other **anew** , and the connection is handled as **the very first one**.
- **Stateless** :- The server will forget everything related to client/browser state
 - This means a HTTP server needs not keep track of any state information.

HTTP Status Code

➤ **Status Code:** it tells the client if the request is succeeded or failed

- 1** : Informational
- 2** : Success
- 3** : Redirection
- 4** : Client Error
- 5** : Server Error

HTTPS:- secured

URI, URL and URN

- **URI (Uniform Resource Identifier):-** is a string of characters used to identify a resource on the internet either by location or name or both

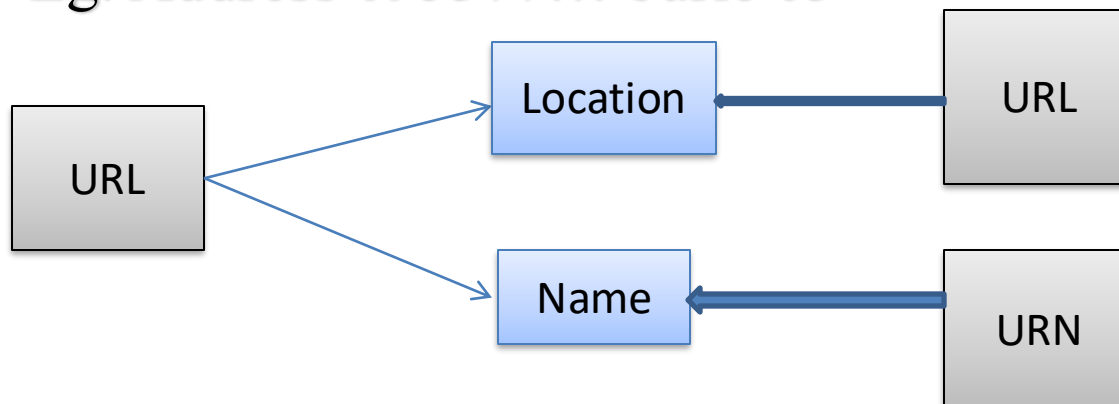
e.g **Name** abebe kebede
 address 095377..... burie 03

- **URN(Uniform Resource Name)** is a string of characters used to identify a resource on the internet by name.

✓ Eg Name: abebe kebede

- **URL (Uniform Resource Locator):-** is a string of characters used to identify a resource on the internet by location

✓ Eg. Address 095377... burie 03



Web Page Addresses (URL)

- Refers to just the address.
- The exact address of a resource on the web
- It is the most used way to locate resources on the web. or
- A URL will have the following format –

[protocol://hostname/other_information](#)

► Or

► `<protocol>://<host>[:<port>][<path>][?<query>]`

☐ E.g. <http://www.somedomain.com/search.php?q=dns&lang=en>

☐ Protocol – identifies the type of protocol to be used for communication

☐ http, ftp, mailto, ...

☐ Host – identifies the machine on which the requested resource is stored

☐ Domain names (eg. www.google.com)

☐ Port – identifies the port number of the web server software on the web server machine

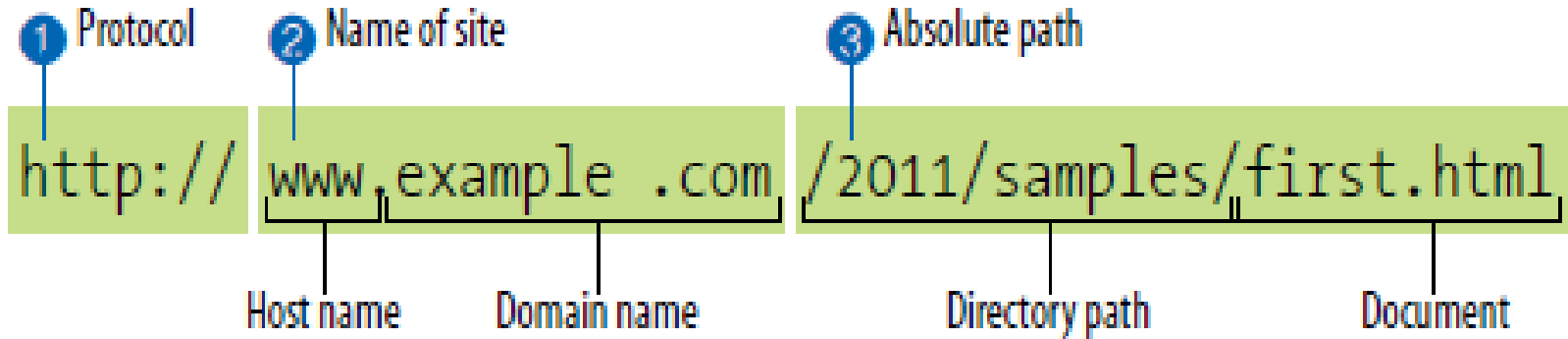
☐ Default port for http: 80

☐ Path – identifies the name and path of the resource on the server

☐ Query – specifies parameters, if any, that should be sent to the server along with the request

☐ has the form: `?var_name1=value1&var_name2=value2&...`

three components of URL



1 **http://**

The first thing the URL does is define the **protocol** that will be used for that particular transaction.

2 **www.example.com:-** identifies the website by its **domain name**.

3 **/2012/samples/first.html**

This is the absolute path through **directories** on the server to the requested HTML document, *first.html*.

Domain Name System (DNS)

- on the internet, DNS automatically converts between the name we type on our web browser address bar to IP addresses of web servers hosting those sites.
- Resolves a human friendly name (eg www.google.com) to a machine friendly IP address (eg 64.233.187.99)
- **Phonebook** of the internet.



Domain Name System



▶ DNS name structure

- ▶ Hierarchical in nature (eg. cs.dmu.edu.et)
 - cs is under dmua (a subdomain of dmua), dmua is under edu, edu is under et.
 - ▶ The highest level is the last component of the DNS address
 - ▶ Labels separated by . (dot)
 - ▶ Labels can be up to 63 characters long and are case insensitive
 - ▶ A maximum of 255 characters is allowed in total
-
- ▶ The last (highest) labels of a DNS name can be:
 - ▶ Three letter code top level domains (TLDs): indicating the type of organization
 - com, edu, gov, net, org, biz, ...
 - ▶ Two letter country codes (CCTLDs): indicating the country
 - et, us, za, uk, tv, ...

► web server vs Web browser

► *At the server:*

- A web server is a **computer that stores a web site**, and is responsible for checking requests for viewing that web site.

- ☐ Apache - open source
- ☐ IIS (Internet Information Services) – Microsoft
- ☐ Squid
- ☐ ...

► *At the client:*

- Web browser : sends/receives requests/responses to/from web servers on behalf of the client and renders content as necessary

- A web browser or Internet browser is a **software application for retrieving, presenting, and traversing** information resources on the **World Wide Web**.
- A web browser allows to view and interact with the web pages on the www
- Example :



- ***Internet Service Provider (ISP)*** :- A company that provides Internet connections to customers.
- **Hypertext**: - Hypertext is a text document that contains links to other text document.
- **Hypertext** allows a user to **move from one web page to another** by using a mouse to click on special **hypertext links**.
- **hypertext** is a text that contains hyperlinks that lead to other documents, usually texts.
- **hyperlink** is a text, word, sentence, button, icon, sometimes a picture, when you hover on it, you will be directed to another web page.

Web page vs website

- **web page** :a document which can be displayed in a web browser also known as pages.
- such documents are written in HTML language.
- web page can be accessed by entering a URL address into a browser's address bar.
- multiple web page make up a website
- a **web site** is a collection of linked web pages and associated resources that share a unique domain name.



The diagram shows the URL `https://www.computerhope.com/jargon/u/url.htm` with brackets underneath identifying its parts: `https` is the Protocol, `www` is the Subdomain, `computerhope.com` is the Domain and domain suffix, `jargon/u` are the Directories, and `url.htm` is the Web page.

- ❖ The website is computerhope.com and the web page is url.htm
- ❖ **Home page**:- the first page of the website

Static vs Dynamic Web Site

Static web site:- a website that contain static web pages.

- a document that, every time it is requested, displays on the browser exactly as it is stored in the server.
- it doesn't change at all, unless if the creator changes it manually.
- the URL address must end with a document extension such as .html,.php... like <http://www.example.com/index.html>

Dynamic website

- website that contain dynamic web pages.
- web pages that are generated using scripting languages and programs intracting with a database on the server side.
- programs that add,restrict,edit or remove data without the creator intervention.
- example: the current date and time at which you are viewing the web page
- the website can be display your profile information every time you log in

frontend and backend Design of website creation.

Frontend design

- “Frontend ”refers to any aspect of the design process that appears in or relates directly to the browser.
- The following tasks are commonly considered to be frontend tasks:
 - ✓ Graphic design and image production
 - ✓ Interface design
 - ✓ Information design as it pertains to the user’s experience of the site
 - ✓ HTML document and style sheet development & JavaScript

Backend design

- “Backend”refers to the programs and scripts that work on the server behind the scenes to make web pages dynamic and interactive.
- The following tasks take place on the backend:
 - ✓ Information design as it pertains to how the information is organized on the server
 - ✓ Forms processing
 - ✓ Database programming
 - ✓ Content management systems
 - ✓ Other server-side web applications using PHP, JSP, Ruby, ASP.NET, Java,

Why Do We Need Websites?

- Websites primarily act as a bridge between one who wants to share information and those who want to consume it.
- If you are running a business, then it is almost imperative for you to have a website to broadcast your offerings and reach out to potential clients at a global stage.
- The following points explain why it is important to have a website:
 - ✓ A website is an online brochure where you can advertise your business offers.
 - ✓ It gives you a platform to reach out to a far-and-wide global customer base.
 - ✓ If you are a blogger, you have the possibility to influence your readers.
 - ✓ You can show all your ideas and publish them on a website.

Website – Domain Name

- A **domain name** is a unique and human-readable address that is used to identify and locate websites on the internet.
- **Domain name** is the part of your online address and your visitors will use it to find you easily.
- **For example**, Tutorialspoint domain name is **tutorialspoint.com**.
- Your domain name is unique to you. Once you have registered it, nobody else can register the same one if you continue to renew it.
- Registering a domain name can be easy because you can select whatever name you want, but what you choose is crucial for your business future or your blogging website.
- some rules that we must keep in mind before we purchase a domain name are

Rule 1 – Choose a domain name that can be your brand name.

Rule 2 – If you have the financial possibilities, then you can purchase domains with **various extensions** such as .com, .net, .info, .biz, etc.

✓ like tutorialspoint.net, tutorialspoint.info, tutorialspoint.biz.

✓ There are domains that are misspelt, such as a similar case is google.com. If you write gogle.com one 'o' is missing, but still it will redirect you to the official website google.com

Rule 3 – You must ensure that this domain name is not a trademark or is copyrighted.

Rule 4 – Find a short domain name.

Rule 5 – Extensions are suffixes, such as .com or .net, at the end of the web addresses. These can have specific use, so ensure to **choose an extension that works for your business.**

Rule 6 – If your domain name is not going by the brand name, then at least you should use specific keywords which describes your business,

Domain Name Extensions

- **First domain type** is Top Level Domain (TLD). These top level domains include any extensions which contain only one suffix .
 - ✓ for example, .net, .info, .biz and so on.
- **Second Level Domains (2LDs)** are domain names containing one more level after the .com or .co suffix.
 - ✓ For example, [.com.au](#) is a second level domain style
- The meanings of some top-level domains are listed as follows:
 - **.com** – commercial
 - **.org** – organization –it is originally intended for use by any non-profit organizations.
 - **.net** – network –It was originally intended for use by domains pointing to a distributed network of computers,
 - **.edu** – education – This TLD is limited to specific higher educational institutions
 - **.mil** – military – The .mil TLD is limited to use by the United States military.

Web hosting & Domain Name Registration

- **Domain name registration** is the process of acquiring a unique domain name for your website.
- Domain names are **registered and managed** through **domain registrars**, which are organizations or companies authorized by the **Internet Corporation for Assigned Names and Numbers (ICANN)**.
- Domain names can be purchased for a specific period, typically ranging from one to ten years, and require renewal to maintain ownership and continue using them
- For example, if you choose a name like "mydomain.com", you should go to a registrar, **pay a registration fee per year for that name**.
- That will give you the right to the domain name for a year.
- You should **renew it annually for the same amount per annum**.
- Some of the biggest registrars where you can register your domain are given below:
 - ✓ GoDaddy - with the URL <https://uk.godaddy.com>
 - ✓ Name – with the URL <https://www.name.com/>
 - ✓ iPage – with URL <https://www.ipage.com>
 - ✓ BlueHost
 - ✓ Hostgator

To register a domain name, you typically follow these steps:

- **Choose a domain registrar:** Select a reputable domain registrar that offers the domain extension (such as .com, .org, .net)
- **Check domain availability:** Use the registrar's search tool to check if the domain name you want is available. If it's already registered by someone else, you'll need to choose a different name or try a different domain extension.
- **Select the desired domain name:** Once you find an available domain name, select it and proceed to the registration process.
- **Provide registration details:** You'll be asked to provide **your contact information**, including your name, address, email address, and phone number.
- **Choose the registration period:** Select the duration for which you want to register the domain name.
- **Complete the registration process:** Follow the instructions provided by the registrar to complete the domain registration process.
- **Domain ownership and management:** Once the registration is complete and your payment is processed, you become the owner of the domain name. **You can manage the domain's settings, such as DNS (Domain Name System) configuration, email forwarding, and other options through the registrar's control panel.**

Web hosting

- **Web hosting** is a service that allows you to publish your website or web application on the internet.
- web hosting service provider, is a business that provides the technologies and services needed for the website or webpage to be viewed in the Internet.
- Web hosting involves renting some space on a physical server where you can store all the files and data necessary for your website to work properly.
- Web hosting involves several key components and features:
- **Storage Space:** provides disk space on a server to store all the files and data related to your website.
- **Server Resources:** allocates server resources such as CPU (Central Processing Unit), RAM (Random Access Memory),
- **Server Management:** handle the technical aspects of server management,
- **Uptime and Reliability:** provide high uptime, which refers to the percentage of time your website is available and accessible to visitors.
- **Domain and DNS Management:** offer domain registration services or integrate with domain registrars.
- **Email Hosting:** (e.g., yourname@example.com).
- **Security:** implement various security measures such as firewalls, SSL certificates, malware scanning, and regular backups to protect websites
- **Technical Support:** provide customer support to assist with server-related issues, troubleshooting, and answering questions related to hosting services.

Web Design Issues

- a) Browser & Operating Systems
- b) Bandwidth and Cache
- c) Display Resolution
- d) Look & Feel
- e) Page Layout and Linking
- f) Locating Information
- g) Making Design user-Centric
- h) Sitemap

a) Browser & Operating Systems

- Web pages are written using different HTML tags and viewed in browser window.
- The **different browsers** and their **versions** greatly affect the way a page is **rendered**, as different browsers sometimes interpret same HTML tag in a **different way**.
- **Different versions** of **HTML** also support different sets of tags.
- The support for different tags also varies across the different browsers and their versions.
- Same browser may work slightly different on different **operating system** and **hardware platform**.
- To make a web page portable, **test it on different browsers** on different operating systems.

b) Bandwidth and Cache

- Users have different **connection speed**, i.e. bandwidth, to access the Web sites.
- Connection speed plays an important role in designing web pages, if user has low bandwidth connection and a web page contains too many images, it takes more time to download.
- Generally, users have **no patience** to **wait** for longer time than **10-15 seconds** and move to other site without looking at contents of your web page.
- Browser provides temporary memory called **cache** to store the graphics.
- When user gives the URL of the web page for the first time, HTML file together with all the graphics files referred in a page is downloaded and displayed.

c) Display Resolution

- Display resolution is another important factor affecting the Web page design, as we do not have any control on display resolution of the monitors on which user views our pages.
- Display or screen resolution is measured in terms of **pixels** and common resolutions are 800 X 600 and 1024 X 786.
- We have three choices for Web page design.
 - Design a web page with **fixed resolution**.
 - Make a **flexible design** using HTML table to fit into different resolution.
 - If the page is displayed on a monitor with a higher resolution, the page is displayed on left hand side and some part on the right-hand side remains blank. We can use centered design to display page properly.
 - Ideally we should use some frameworks for designing like **Bootstrap/Material design**.

d) Look & Feel

- Look and feel of the website decides the overall appearance of the website.
- It includes all the design aspects such as
 - Web site theme
 - Web typography
 - Graphics
 - Visual structure
 - Navigation etc...

e) Page Layout and Linking

- Website contains of **individual web pages** that are linked together using various **navigational links**.
- Page layout defines the visual structure of the page and divides the page area into different parts to present the information of varying importance.
- Page layout allows the designer to distribute the contents on a page such that visitor can view it easily and find necessary details.

f) Locating Information

- Webpage is viewed on a computer screen and the screen can be divided into **five major areas** such as **center, top, right, bottom** and **left** in this particular order.
- The first major area of importance in terms of users viewing pattern is the center, then top, right, bottom and left in this particular order

g) Making Design user-Centric

- It is very difficult for any Web designer to predict the exact behavior of the Web site users.
- However, idea of general behavior of common user helps in making design of the Web site user centric.
- Users either scan the information on the web page to find the section of their interest or read the information to get details.

Planning a Website

- The most important activity in a website development is **planning**.
- To achieve **higher success** of the website in terms of user satisfaction, better planning is needed.
- Before we start developing a website, we should ask question such as
 - **Why** are we developing this website?
 - **What** do we achieve by developing this website?
 - **Who** are the people who will use this website?
 - What are the information **contents**?
 - How are these contents **organized**? What are the **possible ways**?

Effective Navigation

important design element in the web design after page navigation design.

n means the **ways to move from one page to another**

Web site using hyperlinks provided on the page.

on design is not proper then user feels the problem in
ound the pages in your site in a desired manner or gets
and leaves the site.

Tips for Effective Navigation

- Navigation links are either **text based**, i.e. a word or a phrase is used as a link, or **graphical**, i.e. a image, a icon or a logo is used as a link.
- Navigation links should be **clear** and **meaningful**.
- It should be **consistent**.
- Link should be **understandable**.
- Organize the links such that contents are **grouped** logically.
- Provide **search** link, if necessary, usually on top of the page.
- Use **common** links such as ‘about us’ or ‘Contact us’.
- Provide the way to **return** to **first page**.
- Provide the user with **information** regarding **location**
- Horizontal navigation bar can be provided on each page to **directly jump** to any section

End of Chapter one!!!